

Wisdom Favour Chizoba

Manufacturing Systems & Digital Transformation Engineer

wchizoba@gmail.com | +49 157 5591 8419 | 93183, Holzheim a. Forst | github.com/Chizobawisdom



PROFESSIONAL EXPERIENCE

- Oct. 2026 – present **System Developer in Manufacturing (Working Student)**
Schiederwerk GmbH, Nuremberg
- Oct. 2025 – Sept. 2026 **Customer Quality Engineering (Working Student)**
ams OSRAM, Regensburg
- Produced 100+ customer-facing reliability monitoring reports and CDC documentation packages for semiconductor and LED product families, consolidating qualification, reliability, and test data to support customer quality and reliability requirements.
 - Supported PPAP and quality documentation activities through structured capture, validation, and consolidation of reliability and qualification data.
 - Analyzed reliability test results (HTOL, WHTOL, PTC, TC, T&HB, DEW, CDM, and corrosion testing) to identify failure trends, assess root causes, and support customer quality reporting.
 - Collaborated with Customer Quality, Production Quality Engineering, Reliability Engineering, and Manufacturing across global sites to resolve quality issues.
- Apr. 2025 – Oct. 2025 **Quality Systems Engineering (Bachelor's Thesis)**
Mubea Aerostructures GmbH, Munich
- Owned the end-to-end design and implementation of a Microsoft Power Apps solution that transformed Root Cause Analysis (8D, 5 Why, Ishikawa) from a manual process into a standardized digital workflow.
 - Established a structured foundation for quality data by converting unstructured RCA documentation into traceable, analyzable datasets.
 - Improved audit readiness and supported compliance with EN9100 and EASA Part-145/Part-21 through automated documentation and standardized workflows.
 - Optimized quality processes and reduced documentation inefficiencies in aerospace manufacturing.
- Apr. 2024 – Oct. 2025 **Quality Engineering (Working Student)**
Mubea Aerostructures GmbH, Munich
- Supported process planning and quality assurance through data-driven analysis to ensure production performance and process stability.
 - Conducted Gemba walks on the shop floor to verify adherence to quality standards (5S, Standard Work).
 - Contributed to the digitalization of quality documentation and improved data availability.
 - Supported First Article Inspections (FAIs), customer complaints and NCRs, and CAPA processes.
- Oct. 2023 – Apr. 2024 **Data Engineering (Internship)**
ZF Friedrichshafen AG, Friedrichshafen
- Developed management dashboards in Power BI to provide visibility into ZF Supplier Innovations, from submission through implementation, including financial savings.
 - Built data pipelines by cleaning and transforming supplier and process data using Python, Excel, VBA, and SQL, and connected database APIs to Power BI.
 - Delivered monthly and ad hoc analytical insights to support management decisions.
 - Reduced manual reporting effort by automating data pipelines and KPI updates.

PROJECTS

AnyLogic Digital Twin for Warehouse Sequencing & Downtime Minimization

Multi-method simulation with Java agents for dynamic sequencing between high-bay storage and body shop; provides an analytical basis for weighing storage costs against the financial risk of production downtime.

Technologies: AnyLogic · Java

AnyLogic Simulation for Quality Inspection Station Capacity Optimization

Stochastic discrete-event simulation with parameter variation experiments and 95% confidence intervals; demonstrated a 71% reduction in mean cycle time (16.0 → 4.7 min) from adding inspection staff, outperforming an equipment upgrade.

Technologies: AnyLogic · Statistical Analysis

Discrete-Event Digital Twin for Quality Control & Bottleneck Analysis

Python/SimPy simulation of a multi-station production line with stochastic machine breakdowns (MTBF/MTTR), automated rework/scrap logic, and a real-time KPI pipeline (OEE, throughput, bottleneck detection).

Technologies: Python · SimPy

Industrial PLC-to-Cloud OEE Monitor

End-to-end IT/OT architecture: PLC control logic (IEC 61131-3) connected via Modbus TCP to a Python telemetry pipeline and a live-updating OEE dashboard; resolved scan-cycle overruns and counter drift in production.

Technologies: OpenPLC · Python · SQLite · Plotly Dash

Additional projects available at: <https://chizobawisdom.github.io/Portfolio/index.html>

EDUCATION

2027 (ongoing) **M.Eng. Applied AI for Digital Production Management**
Technische Hochschule Deggendorf

2025 **B.Eng. Industrial Engineering and Management**
Technische Hochschule Deggendorf

SKILLS

Industrial Data & AI

- Data analytics & machine learning with Python (pandas, NumPy, scikit-learn), SQL, and predictive modeling for anomaly detection, classification, and forecasting
- Industrial computer vision with OpenCV, TensorFlow, CVAT, and transfer learning for defect detection and image classification
- Data engineering & BI: ETL workflows, data models, KPI pipelines, dashboards, and automated reporting with Python, SQL, Power BI, and Excel

Manufacturing Systems, Quality & Reliability

- Digital twins & simulation with AnyLogic, SimPy, and MATLAB for evaluating queuing behavior, capacity limits, and downtime
- Quality management & continuous improvement: reliability monitoring, PPAP, APQP, RCA, CAPA, 8D, 5 Why, Ishikawa, SPC per IATF 16949, ISO 9001, EN9100

Automation, Integration & Digitalization

- Industrial automation: PLC concepts, Modbus TCP, OPC-UA, MES concepts, real-time monitoring in IT/OT architectures
- Digitalization of manufacturing processes with Power Apps, Power Automate, and VBA to standardize workflows
- System integration: connecting production systems, databases, APIs, and visualization platforms
- Engineering toolchain: SAP, SharePoint, Confluence, Excel, KNIME, MATLAB, LTspice

LANGUAGES

- German – B2 (telc certified)
- English – C2 (native speaker)